

UNIVERSITY OF SINDH JAMSHORO

DEPARTMENT OF COMPUTER SCIENCE

Submitted by:

SANOBER GHOTO

Computer Science Department

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Submitted to: SIR:RAJESH

[Course Name: ARTIFICIAL INTELLIGENCE]

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Assignment 1: AI System Specification

Course: Introduction to Artificial Intelligence

Topic: PEAS Framework and Task Environment Analysis

Chosen AI Application

Medical Imaging Assistant for Early Disease Detection (X-ray Triage)

Task 1: PEAS Specification

Performance Measures:

1. Disease detection accuracy - measured by correct identification of abnormalities in medical images.
2. False positive and false negative rate - measured by avoiding incorrect diagnoses.
3. Response time - measured by how quickly the system analyzes an X-ray.
4. Patient safety improvement - measured by reduction in missed critical cases.

Environment:

The agent operates in hospitals and diagnostic centers where it analyzes medical images such as X-rays. External factors include image quality, patient conditions, different diseases, and availability of clinical data.

Actuators:

- Diagnostic recommendation system
- Alert generation for critical findings
- Report generation module
- Clinical decision support interface

Sensors:

- High-resolution X-ray image input
- Patient medical history data
- Electronic health record information
- Laboratory test results

Task 2: Environment Classification

Fully Observable vs Partially Observable:

Choice: Partially Observable

Justification: The AI cannot observe all patient information or hidden medical conditions only through X-ray images.

Single-agent vs Multi-agent:

Choice: Multi-agent

Justification: The system works alongside doctors, radiologists, and healthcare systems.

Deterministic vs Stochastic:

Choice: Stochastic

Justification: Medical outcomes are uncertain and depend on many unpredictable biological factors.

Episodic vs Sequential:

Choice: Sequential

Justification: Medical decisions may depend on previous patient history and ongoing treatment.

Static vs Dynamic:

Choice: Dynamic

Justification: Patient conditions and available medical information can change over time.

Discrete vs Continuous:

Choice: Continuous

Justification: Medical images and patient health states contain continuous data.

Task 3: Critical Analysis

1. Greatest Technical Challenge:

The biggest challenge is the stochastic nature of the environment because diseases vary between patients and medical data can contain uncertainty. The AI must make accurate decisions while handling incomplete and complex information.

2. Utility Function:

$$U = 0.7 * \text{diagnostic_accuracy} - 0.3 * \text{false_alarm_rate}$$

If the weight of diagnostic accuracy is doubled, the AI will prioritize detecting more possible diseases even if it produces slightly more false alarms.

Structured Representation (JSON)

```
{
  "system_name": "Medical Imaging Assistant",
  "peas": {
    "performance_measures": [
      "disease_detection_accuracy",
      "false_positive_rate",
      "response_time",
      "patient_safety_improvement"
    ],
    "environment": "Hospitals and diagnostic centers analyzing medical images",
    "actuators": [
      "alert_generation",
      "report_generation",
      "clinical_support_interface"
    ],
    "sensors": [
```

```
"high_resolution_X_ray_images",
"patient_history",
"electronic_health_records"
]
},
"environment_classification": {
"observability": {
"choice": "Partially Observable",
"justification": "Hidden medical factors cannot be completely observed from X-rays."
},
"agents": {
"choice": "Multi-agent",
"justification": "Doctors and AI system collaborate in diagnosis."
}
},
"utility_function": "U = 0.7*diagnostic_accuracy - 0.3*false_alarm_rate"
}
```